

Adaptive model predictive control for battery thermal management for electric vehicles

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Abstract— This work focuses on addressing the crucial challenge of battery thermal management in electric vehicles. The goal is to develop an adaptive Model Predictive Control (MPC) strategy that combines predictive power with adaptability to varying dynamics. By formulating the control problem, linearizing the thermal model, and simulating adaptive MPC, the study demonstrates successful temperature control. Simulation results using real driving cycles confirm the controller's effectiveness in controlling the battery temperature. This advancement will contribute to improved battery performance and longevity for future work, thereby enhancing overall vehicle efficiency.

Index Terms—battery thermal management, model predictive control, electric vehicles

I. INTRODUCTION

Thermal management of batteries is a key aspect of electric vehicles (EVs) as it greatly influences the performance, safety, and overall efficiency of the vehicle [1]. Lithium-ion batteries, which are the main energy storage technology used in EVs, are particularly sensitive to changes in temperature [2]. If thermal control is not adequate, it can lead to faster degradation, reduced energy capacity, and potential safety hazards. High temperatures can initiate chemical reactions that damage the battery components, while low temperatures can obstruct ion movement and increase internal resistance [3]. Therefore, it's essential to have efficient thermal management systems in place to maintain batteries within their ideal temperature range, which helps extend their lifespan and ensure stable performance. Moreover, accurate temperature control can enhance the efficiency of charging and discharging processes, thereby maximizing the driving range of EVs and reducing the time required for charging [4].

The process of designing efficient Battery Thermal Management (BTM) systems involves a complex set of challenges that call for creative and novel approaches. These challenges encompass:

- **Nonlinear Battery Behaviour:** Batteries exhibit a nonlinear behaviour in response to temperature variations, which necessitates the use of sophisticated control strategies for accurate prediction and management of these changes. [5].
- **Battery Thermal Dynamics:** The thermal dynamics of batteries function at a slower rate compared to electric drive units, which necessitates longer periods for optimization.

This factor adds layers of complexity and uncertainty to the system. [6].

- **Integration with Vehicle Performance:** The performance of Battery Thermal Management (BTM) systems is closely linked with the overall functioning of the vehicle. Striking an optimal balance between the efficiency and effectiveness of both these elements is of paramount importance. [7].
- **Cooling Efficiency and Energy Reserves:** Effective cooling is crucial for preserving the health of the battery. However, it's important to achieve this without excessively draining the battery's energy reserves, thereby ensuring its optimal performance [8].

To tackle the complex challenges inherent in Battery Thermal Management (BTM) systems, a comprehensive approach is needed. This approach should include advanced control strategies, precise modelling, and a thorough understanding of the dynamic relationship between thermal management and vehicle performance. [9].

Numerous thermal management methods have undergone exploration, encompassing an extensive spectrum of control strategies and technologies. Proportional-Integral (PI) controllers and fuzzy logic methodologies [10], for instance, offer simple yet efficient solutions for maintaining temperatures within desired limits. Meanwhile, the emergence of Model Predictive Control (MPC) has shown its potential by utilizing predictive models to adjust thermal conditions in anticipation of future battery behaviours. [11]. Furthermore, the incorporation of advanced machine learning techniques, such as neural networks [12] and reinforcement learning [13], has proven effective in understanding complex nonlinear thermal dynamics and developing adaptive control strategies to improve battery temperature management.

The aim of this paper is to design an adaptive Model Predictive Control (MPC) controller that is specifically suited for battery thermal management. To achieve this objective, a generic thermal model for the battery with a single equivalent control variable is studied. The optimization task involves managing a multi-input single-output scenario. The inherent nonlinearity of the thermal model is tackled by using a Taylor series linearization technique around the nominal operational point. Moreover, to ensure accuracy and relevance in the control process, the plant model is updated consistently within each interval.

The subsequent sections of the paper are structured as follows: Section II outlines the formulation of the optimization problem. Section III expounds on the process of linearizing the nonlinear plant model. Section IV presents the simulation outcomes. The concluding remarks are provided in Section V.

II. OPTIMIZATION PROBLEM

A. Adaptive MPC

Adaptive MPC is a control method that integrates MPC principles with the ability to adapt to evolving system dynamics [14]. In conventional MPC, a predictive model of the system is utilized to forecast future behaviour and optimize control inputs over a limited time horizon to achieve desired performance. However, in real-world scenarios, system dynamics can change due to various factors like operating conditions, disturbances, or uncertainties in parameters. Adaptive MPC tackles this challenge by continually updating the predictive model in real-time to ensure accurate predictions and control actions, even in the presence of dynamic changes. It continuously monitors actual system responses, updates the predictive model in real time to match real-world behaviour, and subsequently refines control inputs (Fig. 1).

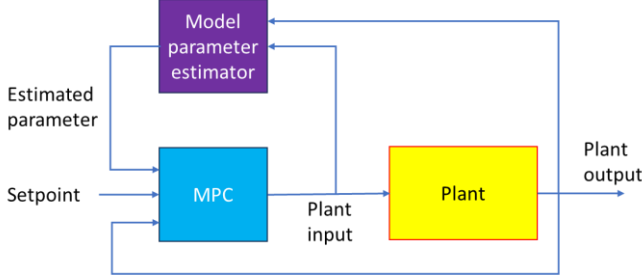


Fig. 1 Principle of adaptive MPC

Because the BTM system is nonlinear, the continuous updating of the predictive model is accomplished by the model parameter estimator after the linearization, which takes the battery thermal and electric models into the calculation by applying the state-space equation given here:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

with

$$x = \begin{bmatrix} T_{bat} \\ SoC \end{bmatrix}, \quad u = \begin{bmatrix} P_{btm_th} \\ T_{amb} \\ P_{trac} \end{bmatrix}$$

and according to the linearization result of the model, the parameters for the system are acquired. Due to the characteristics of adaptive MPC, the parameters of state-space equations for the control system are updated for the accuracy of model prediction.

B. Battery thermal and electric models

To deploy an MPC for a BTM system, it is imperative to incorporate both the electrical and thermal attributes of the battery into the predictive model. Selecting battery temperature T_{bat} and State of Charge (SoC) as the states, they are given by:

$$\begin{cases} \dot{T}_{bat} = \frac{i_{bat}^2 r_{int} + P_{btm_th} + P_{bat_amb}}{m_{bat} C_{th}} \\ \dot{SoC} = -\frac{i_{bat}}{C_{bat}} \end{cases} \quad (1)$$

with

$$i_{bat} = \frac{u_{ocv} - \sqrt{u_{ocv}^2 - 4r_{int}P_{bat}}}{2r_{int}} \quad (2)$$

$$P_{bat} = P_{trac} + \frac{|P_{btm_th}|}{\eta_{btm}} \quad (3)$$

$$P_{bat_amb} = \frac{T_{amb} - T_{bat}}{R_{th}} \quad (4)$$

where i_{bat} and r_{int} are the battery current and internal resistance, P_{btm_th} is the heating/cooling power, P_{bat_amb} is the exchanged power with the ambient, m_{bat} and C_{th} are the battery mass and thermal capacity, u_{ocv} is the open circuit voltage, P_{bat} is the total battery power, P_{trac} is the traction power from the load, T_{amb} is the ambient temperature, R_{th} is the equivalent thermal resistance.

By combining (1)-(4), the states are expressed by:

$$\begin{aligned} \dot{T}_{bat} &= f_{bat}(T_{bat}, SoC, P_{btm_th}, T_{amb}, P_{trac}) \\ &= \frac{\left(u_{ocv} - \sqrt{u_{ocv}^2 - 4r_{int} \left(P_{trac} + \frac{|P_{btm_th}|}{\eta_{btm}} \right)} \right)^2}{4r_{int}} + P_{btm_th} + \frac{T_{amb} - T_{bat}}{R_{th}} \\ &= \frac{\quad}{m_{bat} C_{th}} \end{aligned} \quad (5)$$

$$\begin{aligned} \dot{SoC} &= f_{soc}(T_{bat}, SoC, P_{btm_th}, T_{amb}, P_{trac}) \\ &= -\frac{u_{ocv} - \sqrt{u_{ocv}^2 - 4r_{int} \left(P_{trac} + \frac{|P_{btm_th}|}{\eta_{btm}} \right)}}{2r_{int} C_{bat}} \end{aligned} \quad (6)$$

III. LINEARIZATION OF THE MODEL

The provided equations introduce the control variable for heating/cooling power as P_{btm_th} is the control variable, while T_{amb} and P_{trac} represent measurable disturbances. The system exhibits nonlinearity.

The system is linearized using the Taylor series. Supposing that $(\bar{T}_{bat}, \bar{SoC}, \bar{P}_{btm_th}, \bar{T}_{amb}, \bar{P}_{trac})$ is the nominal operating point:

$$\begin{aligned} T_{bat} &\approx \bar{f}_{bat} - \frac{1}{m_{bat} C_{th} R_{th}} (T_{bat} - \bar{T}_{bat}) + \frac{1}{m_{bat} C_{th} R_{th}} (T_{amb} - \bar{T}_{amb}) \\ &+ \left(\frac{\text{sign}(P_{btm_th})}{m_{bat} C_{th} \eta_{btm}} - \frac{\text{sign}(P_{btm_th}) u_{ocv} \frac{1}{\eta_{btm}}}{m_{bat} C_{th} \sqrt{u_{ocv}^2 - 4r_{int} \left(\bar{P}_{trac} + \frac{|\bar{P}_{btm_th}|}{\eta_{btm}} \right)}} + \frac{1}{m_{bat} C_{th}} \right) (P_{btm_th} - \bar{P}_{btm_th}) \\ &- \left(\frac{1}{m_{bat} C_{th}} - \frac{u_{ocv}}{m_{bat} C_{th} \sqrt{u_{ocv}^2 - 4r_{int} \left(\bar{P}_{trac} + \frac{|\bar{P}_{btm_th}|}{\eta_{btm}} \right)}} \right) (P_{trac} - \bar{P}_{trac}) \end{aligned} \quad (7)$$

And

$$\begin{aligned} \dot{SoC} &\approx \bar{f}_{soc} - \frac{1}{C_{bat} \sqrt{u_{ocv}^2 - 4r_{int} \left(\bar{P}_{trac} + \frac{|\bar{P}_{btm_th}|}{\eta_{btm}} \right)}} (P_{trac} - \bar{P}_{trac}) \\ &+ \frac{\text{sign}(P_{btm_th}) \frac{1}{\eta_{btm}}}{C_{bat} \sqrt{u_{ocv}^2 - 4r_{int} \left(\bar{P}_{trac} + \frac{|\bar{P}_{btm_th}|}{\eta_{btm}} \right)}} (P_{btm_th} - \bar{P}_{btm_th}) \end{aligned} \quad (8)$$

where $\bar{T}_{bat}$, $\bar{SoC}$, $\bar{P}_{btm_th}$, $\bar{T}_{amb}$ and $\bar{P}_{trac}$ are constants at the nominal points. So that (7) and (8) are linear functions of T_{bat} , SoC , P_{btm_th} , T_{amb} and P_{trac} .

IV. SIMULATION RESULTS

A. Feasibility of Adaptive MPC

Common Artemis Driving Cycles Urban is used to test the developed adaptive MPC controller. Assuming an initial battery temperature matching the ambient temperature of 22°C and a target temperature of 25°C, the simulation outcome is depicted in Fig. 2. The battery temperature reaches the setpoint of 25°C after approximately 100 seconds, and subsequently, it remains stable at this temperature. Minor ripples in the temperature profile are evident, attributed to the disturbances stemming from the load. In summary, the designed adaptive MPC controller demonstrates effective temperature control over the battery, achieving the desired outcomes.

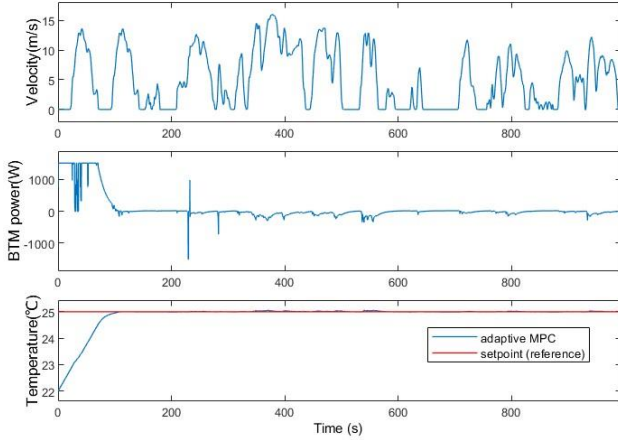


Fig. 2 Simulation results of adaptive MPC (22°C)

B. Comparison with PI controller

To evaluate the performance of the adaptive MPC controller, a PI control system for the same BTM system is built with the inversion-based control for generating the input control command of heating/cooling power $P_{btm,th}$.

The comparisons are done on the performance at the highway cycle of using the two controllers under ambitious scenarios, with the limitation of BTM power being -1500W to 1500W. One is the simulation of a summer environment, keeping the setpoint at 25°C and assuming the environment temperature and battery initial temperature are both 38°C, the performance of adaptive MPC and PI controller is given in Fig. 3 and Fig. 4.

In Fig. 3, the battery temperature is reduced to the setpoint after approximately 500 seconds. And after the stabilization time, the temperature remains stable at the setpoint in the subsequent running time.

In Fig. 4, the battery temperature controlled by the PI controller can be reduced to the setpoint as well. However, the stabilization time is longer than using adaptive MPC, which is approximately 600 seconds. The difference in stabilization time is probably related to the BTM power.

The other scenario is the simulation of a winter environment. The environment temperature is assumed at -20°C, and the initial battery temperature is 20°C, assuming that the battery is pre-conditioned to have better discharge capability. The setpoint is still 25°C. The simulation results are presented in Fig. 5 and Fig. 6.

In Fig. 5, the battery temperature reached the setpoint after approximately 180 seconds. After the stabilization time, the temperature is maintained around 25°C with tiny vibrations.

In Fig. 6, the use of the PI controller managed to increase the temperature after approximately 180 seconds. However, there is a steady-state error exists in the whole process of simulation, with the same PI parameters as applied in the summer environment. The possible reason for the error is the proportional parameter and integral parameter here are not suitable or the limitation of BTM power output.

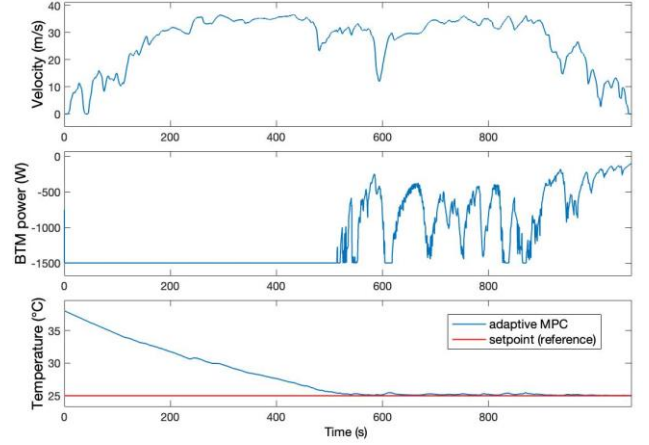


Fig. 3 Simulation results of adaptive MPC (38°C)

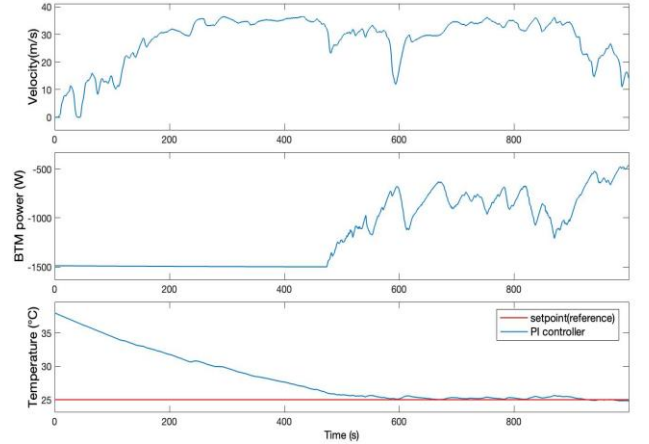


Fig. 4 Simulation results of PI controller (38°C)

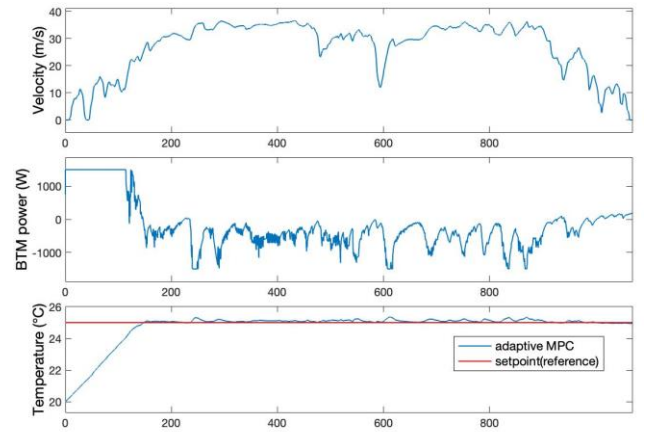


Fig. 5 Simulation results of adaptive MPC (-20°C)

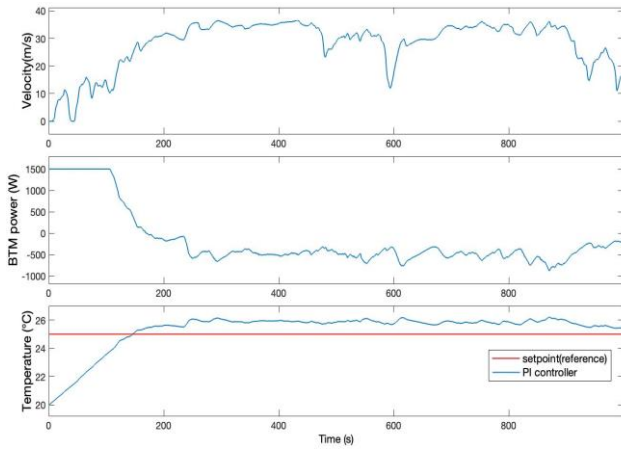


Fig. 6 Simulation results of PI controller (-20°C)

The comparison of the adaptive MPC and PI controller is in the table below.

Table 1 Comparison of Simulation results (38°C)

		Adaptive MPC	PI controller
Stabilization time	Summer (38°C)	500 s	600 s
	Winter (-20°C)	180 s	180 s
consumption of BTM	Summer (38°C)	0.3271KWh	0.3243KWh
	Winter (-20°C)	0.1721KWh	0.1593KWh

In the summer environment, the adaptive MPC reaches the control goals with 0.8% more power consumption than the PI controller. The slight difference in energy consumption is possibly the reason for the shorter stabilization time of adaptive MPC compared to the PI controller. In the winter environment, even though the stabilization times of using two controllers are similar, the model with the PI controller is worse at reaching setpoint temperature. Relatively, the system with adaptive MPC maintains the control results more stably.

CONCLUSION

This study presented the development and application of an Adaptive Model Predictive Control (MPC) strategy for battery thermal management in electric vehicles. The simulation results, based on the implementation of the developed controller using the Common Artemis Driving Cycles urban, showcased the controller's capability to reach and maintain the desired battery temperature setpoint, despite minor ripples caused by load disturbances. Further comparisons are done using the highway cycle. By comparing with a common PI controller, adaptive MPC shows its significance in shorter stabilization time at high temperatures and better control at low temperatures. The successful outcomes demonstrated the potential of the designed adaptive MPC to provide a robust and efficient solution for battery thermal management. This work contributes to advancing the field of battery thermal management control strategies, paving the way for more effective and adaptable approaches in real-world applications.

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